[CEDAR] Data Model

Final Version

1 Data Model

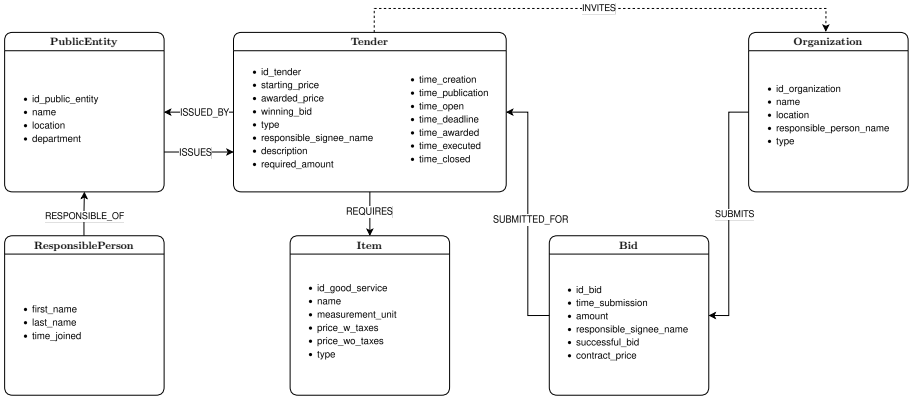


Figure 1: CEDAR Data Model

Figure 1 shows the data model defined for CEDAR MVP. This data model is comprised of 6 entities, 39 attributes, and 7 relations. The main aim of the present document is to describe each of above entities, attributes, and relationships in detail. For that matter entities, their attributes, and the relations between entities will be described in Section 2.

Let it be stated that it is possible that, in the context of the MVP, fields of the data model remain unpopulated in the Knowledge Graph. This is due to the fact that the data model is created to encompass the multitude of use-cases that have been selected towards the MVP, and that leads to some fields not being present in all of them.

2 Entities, Attributes, and Relations

In this section we describe the different entities (alongside with the corresponding attributes) of the CEDAR data model. For each entity we also provide a description of the main relationships that they have with other entities of the model.

2.1 Tender

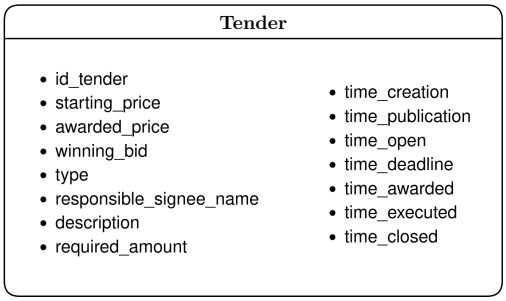


Figure 2: Tender Entity

The entity **Tender** is the central entity of the initial knowledge graph model. A tender represents a formal request to supply goods or provide a task, service, or job. For a tender, different organizations will submit bids including an offered price. The entity is depicted in Figure 2.

This entity is characterized by the attributes *id_tender*, *starting_price*, *awarded_price*, *winning_bid*, *type*, *responsible_signee_name*, *description*, *required_amount*, *time_creation*, *time_publication*, *time_open*, *time_deadline*, *time_awarded*, *time_executed*, and *time_closed*.

- *id_tender* is an unique identification code for the tender, assigned by the organization issuing it. This code will be comprised of an undetermined number of alphanumeric characters.
- *starting_price* is the initial price set when creating the Tender to provide a reference for the submitted bids.
- *awarded_price* is the price that will be paid by the winning bid.
- *winning_bid* is the unique identification code for the bid submitted by an organization which will supply the services or goods required for the tender. This code will be comprised of an undetermined number of alphanumeric characters.
- *type* is the characterization of the tender. This characterization can be with respect to what is required to be supplied (such as "goods", "services", "works", or any other possibility), but also with respect to how the

tender needs to be performed (such as "negotiated", "urgent", "open", or any other possibility). In the case that multiple types are required, these should appear concatenated.

- *responsible_signee_name* is the full name of the person within the public entity that is responsible for the publication, fairness, and well-being of the tender. It will be comprised of an undetermined number of alphabetical characters.
- *description* is a detailed listing and explanations of the goods and services required by the tender, as well as of the context on which the tender happens. It will be comprised of an undetermined number of alphanumeric characters.
- *required_amount* is the number of items that are required by the Tender. It will be comprised of a number.
- *time_creation* is the date and time of the creation of the tender by the public entity. This attribute will have a datetime type.
- *time_publication* is the date and time of the publication of the tender by the public entity. This attribute will have a datetime type.
- *time_open* is the date and time on which the tender is open. This attribute will have a datetime type.
- *time_deadline* is the date and time of the end of the tender, the deadline for submitting any bids to the tender. This attribute will have a datetime type.
- *time_awarded* is the date and time on which the tender is awarded to specific bid. This attribute will have a datetime type.
- *time_executed* is the date and time on which the tender gets executed by the organization that submitted the succesful bid. This attribute will have a datetime type.
- *time_closed* is the date and time on which the tender gets closed after all the corresponding goods and services have been provided. This attribute will have a datetime type.

2.1.1 Tender relationships

Tenders have four types of relationships with other types of entities from the model, in order to capture the underlying authorities, goods or services that must be supplied, the received bids, and possible invitations to certain organizations. We briefly describe each relation in the following lines.

First, every tender has a one-to-one relationship with a **PublicEntity**. The Public Entity provides details about the organization that issues the Tender. The relationship is characterized by the *id_tender* and *id_public_entity* properties of each Entity. This is depicted in Figure 3.

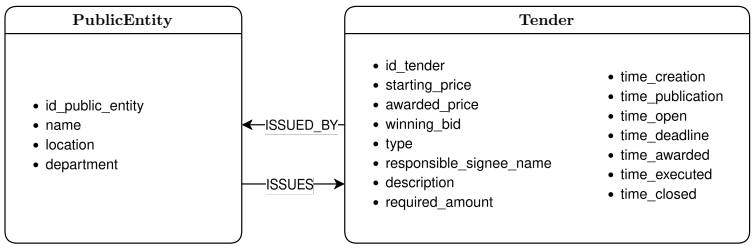


Figure 3: PublicEntity - Tender Relation

Second, Tenders have a relation with **Item** entities. This relation details the contents required by the tender in the form of a number of items, be they goods or services. In the initial version for the MVP a tender will only require one type of item (with multiple relationships being possible for more complex Tender definitions). The relationship is characterized by the *id_tender* and *id_good_service* properties of each Entity. This is depicted in Figure 4.

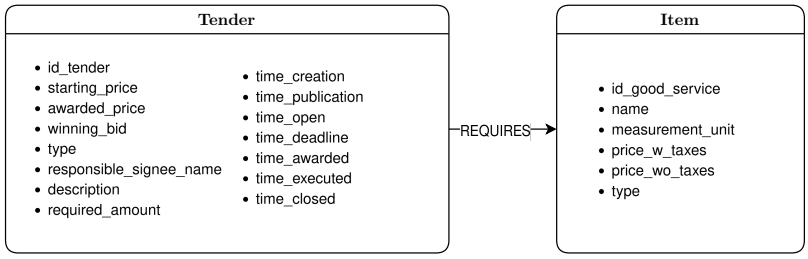


Figure 4: Tender - Item Relation

Third, Tenders have a relation with one or more **Bids**. The relation links each Bid to the corresponding Tender. In particular, any tender is expected to receive a number of bids that do their best to fulfill the needs of the tender. The relationship is characterized by the *id_tender* and *id_bid* properties of each Entity. This is depicted in Figure 5.

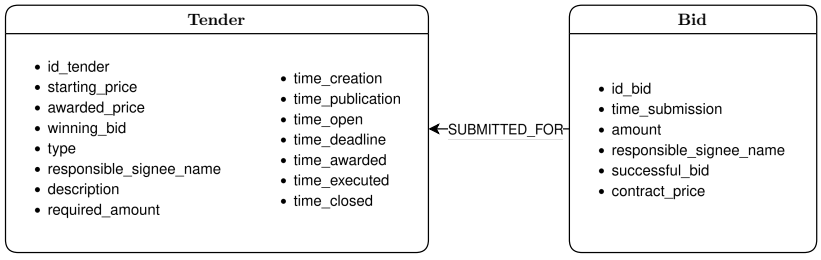


Figure 5: Tender - Bid Relation

The fourth and final relation is established between Tender and **Organi-**

zation. The relation describes the possibility that a tender invites a specific organization to participate in it. This organization in its turn can participate or not on the tender. The relationship is characterized by both, the *type* properties of each entity and the *id_tender* attribute. This is depicted in Figure 6.

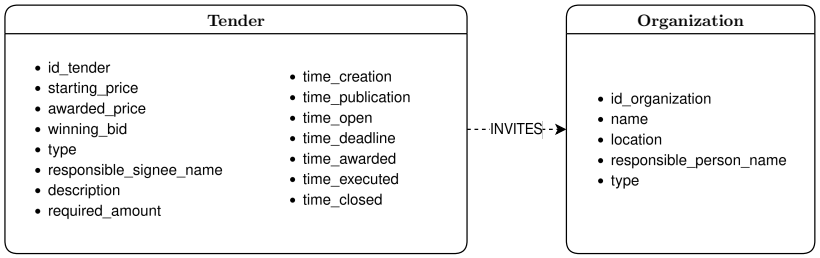


Figure 6: Tender - Organization Relation

2.2 Public Entity

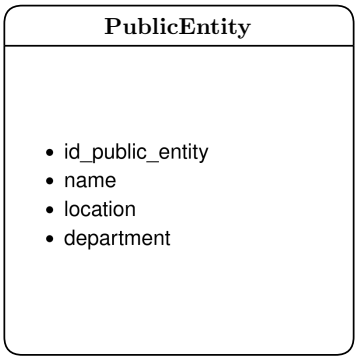


Figure 7: PublicEntity Entity

The entity **PublicEntity** is the representation of an organization or body providing services to the public on behalf of the government. In the case of CEDAR it will be the organization that is issuing the tender on which later on the participants will bid. The entity is depicted in Figure 7.

This entity is characterized by the attributes *id_public_entity*, *name*, *location*, and *department*.

- *id_public_entity* is an unique identification code for the organization. This code will be comprised of an undetermined number of alphanumeric characters.
- *name* is the name of the public entity that issues the tender. This name might not be unique. It will be comprised of an undetermined number of alphanumeric characters.

- *location* is the address in which the headquarters of the public entity are located. It will be comprised of an undetermined number of alphanumeric characters.

- *department* is the suborganization within the public entity that is issuing the public tender. It will be comprised of an undetermined number of alphanumeric characters.

2.3 Item

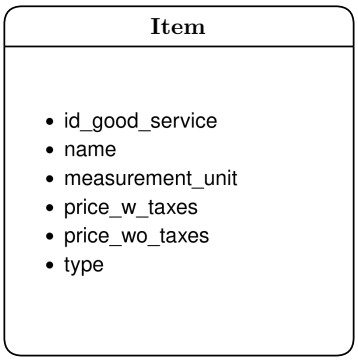


Figure 8: Item Entity

The entity **Item** is the representation of a request for one type of goods or services within a Tender. The entity is depicted in Figure 8.

This entity is characterized by having the attributes *id_good_service*, *name*, *measurement_unit*, *price_w_taxes*, *price_wo_taxes*, and *type*.

- *id_good_service* is the unique identification code of the good/service that is expected to be supplied for the tender. This code will be comprised of an undetermined number of alphanumeric characters.

- *name* is the name of the item that is required for the bid. This name might not be unique. It will be comprised of an undetermined number of alphanumeric characters.

- *measurement_unit* is the definite magnitude used to quantify or measure the required item. It will be comprised of an undetermined number of alphanumeric characters.

- *price_w_taxes* is the cost of one item including all the corresponding taxes, and expressed in euros. It will be comprised of a number.

- *price_wo_taxes* is the cost of one item without including any of the corresponding taxes, and expressed in euros. It will be comprised of a number.

- *type* is specification of the kind of item that will be required for the tender: either an item or a good. It will be comprised of an undetermined number of alphabetic characters.

2.4 Bid



Figure 9: Bid Entity

The entity **Bid** is the representation of the offer that one organization makes to fulfill the requirements of the tender issued by the public entity. In particular, this bid will offer a certain amount of the services/goods needed by the original tender by a certain price. The entity is depicted in Figure 9.

This entity is characterized by having the attributes *id_bid*, *time_submission*, *amount*, *responsible_signee_name*, *succesful_bid*, and *contract_price*.

- *id_bid* is the unique identification code for the bid, submitted by an organization which will supply the services or goods required for the tender. This code will be comprised of an undetermined number of alphanumeric characters.

- *time_submission* is the date and time of the submission of the bid by the organization. This attribute will have a datetime type.

- *amount* is the number of goods/hours of service, that the bid is contributing to those expected by the tender. It is possible that this differs from the required by the tender. It will be comprised of a number.

- *responsible_signee_name* is the full name of the person, within the organization that submits the bid, that is responsible for the publication, fairness, and well-being of it. It will be comprised of an undetermined number of alphabetical characters.

- *successful_bid* is the indicator of the success of the bid in winning the tender. It will be a boolean stating true if the organization did win the tender, and false otherwise.

- *contract_price* is the monetary amount that will be charged to the public entity for the provision of the goods/services required expressed in euros.

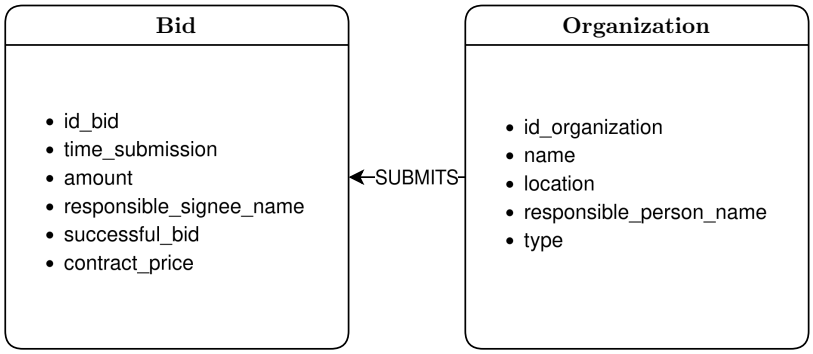


Figure 10: Bid - Organization Relation

There is a one to one relation between and Bid and Organization. This allows to relate information about the organization that submits a specific bid for the tender that the Public Entity has issued. It is expected that an organization will only issue one bid for the same tender, but multiple bids aimed at different tenders.

The relationship is characterized by the *id_bid* and *id_organization* properties of each Entity.

2.5 Organization

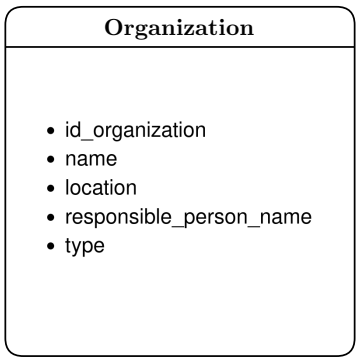


Figure 11: Organization Entity

The entity **Organization** is the representation of the entity that submits the bid for the tender issued by a public entity. This organization is aimed to be a private company, or a conglomerate of them. The entity is depicted in Figure 11.

This entity is characterized by having the attributes *id_Organization*, *name*, *location*, *responsible_person_name*, and *type*.

- *id_organization* is an unique identification code for the organization. This code will be comprised of an undetermined number of alphanumeric characters.
- *name* is the name of the organization that submits the bid. This name might not be unique. It will be comprised of an undetermined number of alphanumeric characters.
- *location* is the address in which the headquarters of the organization are located. It will be comprised of an undetermined number of alphanumeric characters.
- *responsible_person_name* is the full name of the person within the organization that authorizes bids. It will be comprised of an undetermined number of alphabetical characters.
- *type* is the kind of organization that submits the bit. It could be a private company, a conglomerate, or any other type of organization that might be able to submit a bid. It will be comprised of an undetermined number of alphabetical characters.

2.6 Responsible Person



Figure 12: ResponsiblePerson Entity

The entity **ResponsiblePerson** is the representation of the person responsible of a specific public entity. In the case of CEDAR it will be the person that holds the highest level of responsibility within the public entity that is issuing the tender on which later on the participants will bid. The entity is depicted in Figure 12.

This entity is characterized by the attributes *first_name*, *last_name*, and *time_joined*.

- *first_name* is the name of the first name of the person responsible for the public entity that issues the tender. This name might not be unique. It will be comprised of an undetermined number of alphanumeric characters.
- *last_name* is the name of the last name of the person responsible for the public entity that issues the tender. This name might not be unique. It will be comprised of an undetermined number of alphanumeric characters.
- *time_joined* is the date and time on which the person acquired the responsibility over the public entity. This attribute will have a datetime type. In case a model instance contains multiple **ResponsiblePerson** entities for the same Organization, the active **ResponsiblePerson** at a given time will be the one whose *time_joined* more closely precedes that given time.

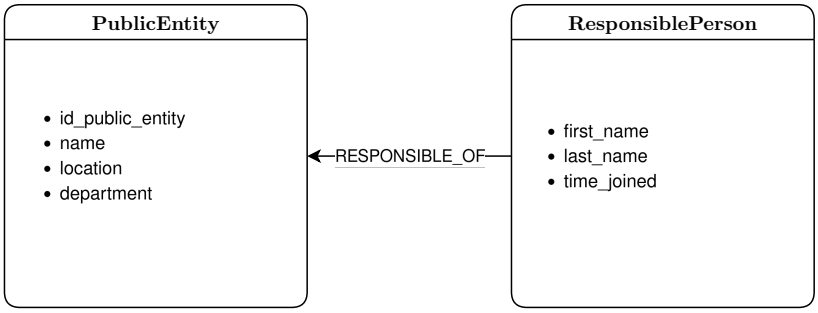


Figure 13: ResponsiblePerson - PublicEntity Relation

There is a one to one relation between and Bid and Organization. This allows to relate information about the organization that submits a specific bid for the tender that the Public Entity has issued. It is expected that an organization will only issue one bid for the same tender, but multiple bids aimed at different tenders. This is depicted in Figure 13.

The relationship is characterized by the *id_bid* and *id_organization* properties of each Entity.

3 Indicators

There are a number of possible indicators of a lack of transparency that were defined by the Italian, Slovenian, and Ukrainian pilots. These indicators are defined over the data model as operations in such a way that they can be calculated easily. These indicators are part of those defined in D5.1. Demonstration Preparation Report v1.0 and there the full explanation of each of them can be found.

3.1 Italian Pilot

In the Italian Pilot UC#2 Negotiated procedures are specifically scrutinized. These procedures can also be started for three different kinds of assets: works, goods, services. Therefore, Negotiated procedures will be modelled as **Tender** entities with the attribute *type* specified as "negotiated" joint with one of the three kinds ("works", "goods", "services").

UC#2-AI#1: NUMBER OF Negotiated procedures for single internal unit.

> This indicator is calculated by measuring the number of ISSUES relations where the same **PublicEntity** entity (as determined by the id) is connected to **Tender** entity with the attribute *type* specified as "negotiated". Different internal units will have unique **id_tender**.

UC#2-AI#2: NUMBER OF Negotiated procedures for item category.

> This indicator is calculated by measuring the number of ISSUES relations of a **PublicEntity** entity connected to **Tender** entity with the attribute *type* specified as one of the three kinds of "negotiated". *work, good, service.*

UC#2-AI#3: List of invited companies and received bids.

> This indicator is calculated by calculating the ratio of number of SUBMITS relations of **Organization** with *type* set to "invited" that connect to a **Bid** that are SUBMITTED_FOR for the same **Tender** with the same *id_tender*, and the number of SUBMITS relations of **Organization** with *type* set to any other value that connect to a **Bid** that are SUBMITTED_FOR for the same **Tender** with the same *id_tender*. Note: the data model does not explicitly capture the concept of invited organization to submit a bid, so this would be a possible workaround by making use of the *type*. Another alternative would be to include the full list of invited organizations as part of the Description (or even their number), which is already free text, and use that to compute the ratio rather than modeling relationships that express invitation and not submission.

3.2 Slovenian Pilot

UC#5-AI#1: NUMBER OF BIDS: (Relative) number of submitted bids for a specific good / service in historical data.

> This indicator is calculated by first selecting a specific **Item** (characterized by the *id_good_service*). Then, for each **Tender** that has a REQUIRES relationship with the entity, measuring the number of SUBMITTED_FOR relations connecting **Bid** entities to the **Tender**. From there statistical analysis per unique item can be performed.

UC#5-AI#2: NUMBER OF BIDS: (Absolute) number of submitted bids for a specific tender.

> This indicator is calculated by measuring the number of SUBMITTED_FOR relations connecting **Bid** to **Tender** with the same *id_tender*.

UC#5-AI#3: REQUIREMENTS: Too strict / limiting / specific requirements can be defined to exclude other bidders.

> This indicator is calculated by researching the text included in *description* of all the **Tender** with the same *type*.

3.3 Ukrainian Pilot

UC#2-AI#5: The ultimate beneficial owner (controller) of the legal entity is its chief executive or signatory

> This indicator is calculated by extracting the *responsible_person_name* of the **Organization**

UC#5-AI#1: Rebuild Ukraine procurement: winning bids

> This indicator is calculated by extracting the *winning_bid* of **Tender** related to the SUBMITTTED_FOR **Bid**.

UC#5-AI#4: Rebuild Ukraine procurement: Bidders connected to one company only

> This indicator is calculated by ensuring that all **Organization** only have one relation SUBMITS per **Bid** SUBMITTED_FOR a **Tender**.